

Predictive Modeling for Alzheimer's Disease: A Survey of Machine Learning and Deep Learning Techniques

Sharvari J N¹, Revathi V²

¹Research Scholar, Department of CSE, School of Engineering, Dayananda Sagar University, Bengaluru

²Associate Professor, Department of CSE, School of Engineering, Dayananda Sagar University, Bengaluru

Email id: sharvari.jn.res-soe-cse@dsu.edu.in, revathi.v-cse@dsu.edu.in

Abstract- Alzheimer's Disease (AD) is a progressive chronic neurodegenerative disease. It is a brain disorder which is categorized by a decline in thinking ability, memory power and performing day to day tasks in patients suffering from AD. The time of diagnosis plays a very crucial role in preventing the human brain from the effects that AD has on brain tissues, behaviour, memories and neurons that impair the senses. Even though AD has no remedy, prediction and early diagnosis can help in reduction of development of the disease. Magnetic Resonance Imaging (MRI) is used as a tool with the scanned brain images for visualizing changes associated with AD. Different approaches for early detection and prediction of AD have been proposed and this work has summarized the various ML and DL techniques that help in contributing the same. Various classifiers used in identifying AD have been explored and the necessary improvements required for enhancing the performance are also discussed in the paper.

Keywords: *Alzheimer's Disease (AD), Magnetic Resonance Imaging (MRI), Mild Cognitive Impairment (MCI), Principal Component Analysis (PCA), Machine Learning (ML) and Deep Learning (DL).*

I. INTRODUCTION

AD is a chronic progressive neurodegenerative disease with symptoms such as gradual memory loss and decline in cognitive skills of the human brain causing deterioration in thinking abilities. AD is often diagnosed in patients in their mid-60's. It is usually caused by neuronal cells dying due to two irregular protein particles namely clumps and pests. Humans suffering from AD initially start with small symptoms such as minimal adjustments in behavior, trouble searching for words and eventually facing mortality in day-to-day activities. The performance of the human brain decreases significantly and gradually over time.

MRI is one technique through which the changes in patients suffering from AD can be monitored. The progression of the disease can be slowed down by early detection and diagnosis thus reducing the expenses incurred for treating the same.

Alzheimer's disease can be categorized into several stages beginning with Mild Cognitive Impairment (MCI)[1], progress Mild Cognitive Impairment (pMCI) and stable. Mild Cognitive Impairment (sMCI). The age of the patient, level of methylation in the DNA are some of the factors that help in identifying AD. One of the techniques for predicting AD is by using Long Short-Term Memory (LSTM) a neural network [2, 3] that is recurrent [4] and could use past information to connect to the current task. Another method of predicting the disease[5] is by using Ordered Core Features (OCF) which reveals the connectivity of the functional human brain and its relationships with different categories of AD[6]. Prediction of AD [7] can also be done using Machine Learning Techniques[8] using Principal Component Analysis (PCA) and Artificial Intelligence techniques [9, 10] and biomedical methods[11] such as DNA Methylation and genetic analysis. Transfer learning techniques and their architectures can also be integrated with other methods related to Computer vision and their types which help in recognition of images effectively.

II. RELATED WORK

This section presents some of the recent methodologies documented in the area of prediction and detection of AD.

Xin Hong. et al [12] have suggested a method to predict AD using LSTM. This technique uses both

activation layers and fully connected layers to build and encode a connection with the next upcoming stage of AD and the current existing features. Studies show that AD would have begun 20 years before the patient shows symptoms and real associations with changes in the brain. But LSTM is capable of identifying dependencies based on a repetitive neural network model. Time step data is used based on a data pipeline that is preprocessed instead of time series data which disrupts the accuracy in prediction. A time sequence model is predicted based on the data received. Comparisons are made with the efficiency of the current algorithm with other ML algorithms and they are tested for stability and sensibility based on the size of the data along with different features. This paper used a data set from AD Neuroimaging Initiative (ADNI) data base wherein the images were extracted which contained about 1000 cases that showed progress in AD. The preprocessing stages included skull stripping, registration, segmentation, normalizing the images and smoothening. The processing of the image is done by following a procedure which includes stripping the images from the skull, registering the images with one another and performing segmentation of the main brain structures. These are then normalized and smoothened to a standard size. MRI technique is used to capture the internal structure of the brain. The atrophy between the scans is measured by the difference in loss of volumes in a specific region of the brain. It is understood that the sequence dependent temporal context has prediction abilities based on different brain Regions of Interests (ROIs). LSTM being a special kind of neural network can selectively be able to retain past information and current task. Therefore, a fully connected LSTM network has layers within it to bring out the relation between its present status and the forthcoming stage of the disease. The degree of damage to neurons in different parts of the brain is reflected based on the symptoms and the person by using a cross-validation technique. This method used time relative biomarkers with the disease status. This paper concluded by notifying that the Cortical Thickness Average (TA) was a prominent characteristic to predict growth of AD when compared to other biomarkers.

Gerardo Alfonso Perez and Javier Cabellero[13] proposed a paper that identifies Alzheimer's disease through DNA methylation and Artificial Intelligence Techniques. This paper focused on identifying and

developing important biomarkers for the disease. With data that is available publicly and the possibility of estimating methylation levels of CpGs in DNA a lot of information is generated. Addition of methyl groups to existing CpGs get them methylated. The measure of methylation in the DNA is an important biomarker for several illnesses as well as to approximately predict the age of the patient. A variety of tasks that are used to predict and classify the disease are found in the field of engineering along with Artificial Intelligence techniques. A methylation level 0 indicates that a DNA is completely unmethylated and level 1 indicates 100% methylated. They proposed an algorithm in such a way that multiple data sets could be extracted. It contained two steps which included initial filtering and the main algorithm. The aim of initial filtering was it reduced the dimensionality of the problem. The main algorithm comprised of vector grid creation, mapping, filtering and generating Hit Ratio and find two highest successive rates. The steps were repeated until a desired Hit Rate was achieved. A considerable reduction in the number of CpGs analyzed was observed which increased the precision of classification than when all CpGs were used. The paper concluded by claiming that identifying techniques that ascertain whether an individual has the disease is of utmost importance. The approach followed an out of sample technique by using a multi data set approach which was a significant milestone. The results proved that when approximately 4000 CpGs were used the specificity and sensitivity values were at 0.9485 and 0.9007 respectively and also claimed for higher Positive Predict Value (PPV) and Negative Predict Value (NPV). The paper finally ended by concluding that as data sets become available based on the blood group, the algorithm could be applied on them to become an additional tool for diagnosis of the disease.

M Sudharsan and G Thailambal[14] proposed a paper that used Principal Component Analysis and ML techniques for prediction of Alzheimer's disease. This paper used MCI, sMR, IVM, RELM and SVM techniques for diagnosing Alzheimer's disease. Approximately data of 200 individuals between the age of 65 and 96 were collected and used. The steps in the methodology followed were preprocessing of the data, structuring the details, classification of stages of the progression of AD, selection of features, using classifiers such as IVM, SVM, RELM. The preprocessing stage consists of

modification of movement, picture tracking, skull stripping. This paper used greedy score-based strategy to choose essential vectors based on the functionality. To handle dynamic data transformations a distinct method based on kernels was also taken. The performance comparisons were made based on data from ADNI repository and the study revealed that by using RELM the accuracy for classifying AD increased. The paper also concluded by mentioning that RELM was the technique that considerably improved accuracy in the case of binary and multiclass classification methods. This paper mainly focused on designing techniques that could identify progression of the disease and would further experiment with improving enhancement of accuracy in the future.

Jinhua Sheng et al[15] proposed a paper for identifying and characterizing different stages toward Alzheimer's disease using ordered core features and machine learning. In this paper, the authors use the Ordered Core Feature concept to identify the functional connectivity in the brain and their relationship among the varying forms of AD, EMCI, LMCI and HC. SVM classifiers were created by extracting a progressive relationship and the use of which significantly changed a set of ordered core features. The methodology comprised of data preprocessing and parcellation, construction of networks based on connectivity, measuring the brain network, analyzing the variability and a classifier such as SVM. Preprocessing was done using data acquired from ADNI2 by parcellation method where raw data from DICOM was converted and registered in the HCPMMP over around 360 areas. The correlation coefficients between any two areas in the

brain is computed using a dense matrix. Connections that were unreal or those that did not make any real connection were removed from the network based on the dPSW value. For each subject in the group the optimal value is computed. Pearson coefficients were computed to understand the association between the regions of the brain on the areas parcellated. The connectivity was represented by a matrix where self-connected nodes were marked with zero (diagonal elements).

Studies showed that the measures of the brain network which were graph based had a better impact in revealing the connections in the human brain. Depending on the networks (weighted/ binary) different local and global measures were analyzed for various subjects and then used them as a feature of machine learning to distinguish between AD patients and Healthy Controls.

The concept of OCF was used in the variability analysis step to differentiate among the different stages of AD. The authors used SVM and multiple other classifiers to perform the study. The performance of the classifiers was then computed based on the four indicators False Positive, False Negative, True Positive and True Negative. The paper concluded by saying that the significant changes could be measured by the network in a set of areas and those changes were in a specific order of HC, EMCI, LMCI and AD respectively or reverse. The table 1 provides a comparison of the different ML algorithms that can be used for prediction of AD with their pros and cons.

TABLE I. Comparison of Different ML Algorithms for Prediction of AD

Algorithm	Characteristic/Feature	Pros	Cons
Logistic Regression (LR)	Identifies models based on certain features	Easy to interpret	Cannot handle non-linear data
Support Vector Machine (SVM)	Categorizes patients based on hyperplanes	Performs better on smaller amounts of data with higher dimensions	Difficult to achieve optimal performance measures
K-Nearest Neighbors (KNN)	Classifies based on neighbors and their similarity	Straightforward	Difficult to perform computation as the data set/size increases
Convolutional Neural Networks (CNNs)	Understands spatial patterns	Works best with MRI scanned images	High computational cost for huge data required
Recurrent Neural Networks (RNN, LSTM)	Takes into consideration patterns related to time and sequential data	Good for prediction with progression than at current state	Requires a huge data set
Principal Component Analysis (PCA)	Lessens the dimension before classification	Better for data with higher dimensions	Possibility of losing information

Algorithm	Characteristic/Feature	Pros	Cons
Random Forest (RF)	Based on a group of decision trees	Can handle non-linear data and noise	Difficult to interpret
Hybrid Models (e.g., Deep Learning + Feature Selection)	Combination of ML and DL methods	High performance achieved	Comparatively complex

III. MOTIVATION

AD has no particular cause for occurrence. But there has been a drastic rise in number of people affected by it and studies show that the count is only going to increase further. The changes in progression of patients from a state of being healthy to suffering from AD is widely spread over many years. The graph (figure 1) below indicates the rising count of the number of people (in millions) who will be diagnosed with AD till the year 2050. This alarming number certainly needs attention to find ways to detect and predict the disease at an early stage in order to avoid the increased risks of late-stage diagnosis that leads to significant cognitive impairment.

The table 2 contains the details of the projected global cases of AD by the year 2050.

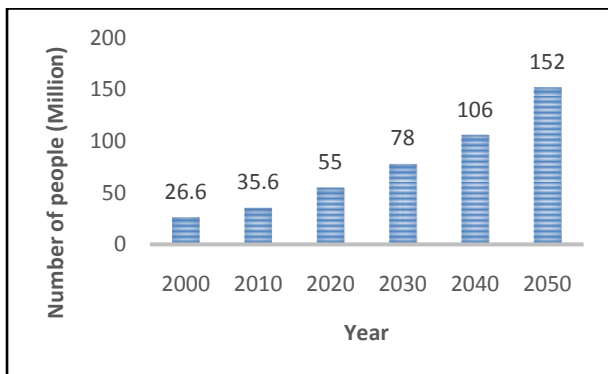


Fig 1. Projected Global Alzheimer's Cases till 2050

TABLE II. Projected Global Alzheimer's Cases Till 2050

Year	No. of people (in millions)
2000	26.6
2010	35.6
2020	55
2030	78
2040	106
2050	152

AD being one of the major existing neuro disorders all over the world, the prediction and detection of the disease using ML at the earliest has truly emerged as

one of the most crucial areas under research. The most important thing to consider is identifying the disease at early stages unlike the conventional clinical approaches thus reducing the complications that involve therapies, hospitalizations etc. The methods used by ML leverage different types of data such as images, genetic data and assessments done by clinicians. The models that can be used for predicting help in identifying individuals who are at a high risk of succumbing to the disease encouraging preventive measures and interventions at the right time. Combining prediction and detection enhances the precision and also reduces the stigma in the society with respect to the disease.

IV. MACHINE LEARNING FOR ALZHEIMER'S DISEASE PREDICTION& DETECTION

Machine Learning and its techniques have become an integral part of medical applications specifically in helping with prediction and detection of AD. This helps in identifying and differentiating the various patterns linked with the disease. The ML models help in early intervention of detecting the disease thus reducing the risk of progression of the disease. Although ML models exhibit great efficiency in varied applications in the field of medicine, it has its own challenges when applied in real life health care scenarios. Therefore, analyzing the different models and suitably choosing the most appropriate model yields accurate results. Although there have been multiple techniques used to predict and detect AD, there still lies a gap in doing that at an earlier stage of a patient's life. Neural network models can enable prediction and detection of the disease by analyzing the complexity and different categories of data such as MRI images, scans and information related to genetics etc.

Role of Machine Learning in Alzheimer's Detection

Machine learning can analyze various types of data to help detect or predict Alzheimer's disease, including:

Neuroimaging Data: MRI and PET scans of the brain can be analyzed using ML models to identify

patterns linked to Alzheimer's. Changes in brain structure or metabolism that may not be visible to the human eye can be detected through computational methods.

Genetic Data: Certain genetic markers are often related with a higher risk of developing Alzheimer's. Machine learning can analyze genetic data alongside other risk factors to assess the likelihood of disease onset.

Cognitive Test Results: Machine learning can help track changes in cognitive performance over time. Even subtle declines in memory, attention, or problem-solving abilities can be flagged by algorithms.

Electronic Health Records (EHR): ML models can process a large amount of patient data from records that are maintained electronically which include medical data with respect to previous hospital admissions, prescribed medical drugs, laboratory reports, and clinical notes, to find correlations that may suggest the development of Alzheimer's.

V PROPOSED METHODOLOGY

In predicting AD, it is still not possible for us to state that one among the many methods discussed in the literature survey is generically better than the other since each of them have their own approach. LSTM can be considered better when we want to take into account certain parameters such as changes in MRI, decline in cognitive abilities and the frequency of a patient visiting the hospital. LSTM can also be used to monitor the progression of the disease than just its current state. DNA methylation can be claimed to be better when early diagnosis is to be done than later when the symptoms are evident. It is also associated with a biological biomarker correlated to the environment or aging related issues. PCA can be considered as a more generalized and practical approach to deal with high dimensional sensitive data such as genetic data and MRI voxels. The OCF technique when combined with LSTM, DNA methylation and PCA enhances and improves the accuracy thereby helping in predicting which features matter the most. The figure 2 below depicts the flow of the methods that facilitate prediction of Alzheimer's disease.

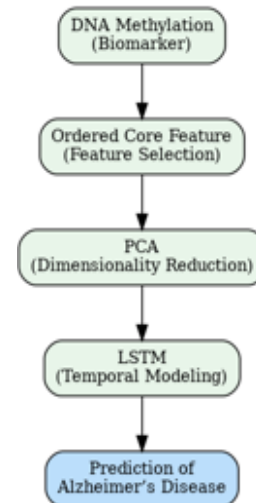


Fig. 2: Flowchart showing prediction of Alzheimer's Disease

VI RESULTS AND DISCUSSION

The performance was measured using various metrics for the different Machine Learning Algorithms and are shown in the table 3 below. It is found that the accuracy is highest when a combination of the different ML algorithms is used. The figure 3 shows the performance comparison of the AD prediction methods taking into consideration the accuracy and the AUC.

TABLE III. Performance metrics of ML methods

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC (%)
LSTM	84.2	82.5	83.1	82.8	86.4
DNA Methylation	80.7	78.2	79	78.6	83.5
PCA	77.9	76.8	75.5	76.1	80.2
Hybrid	89.6	87.8	88.5	88.1	91.3

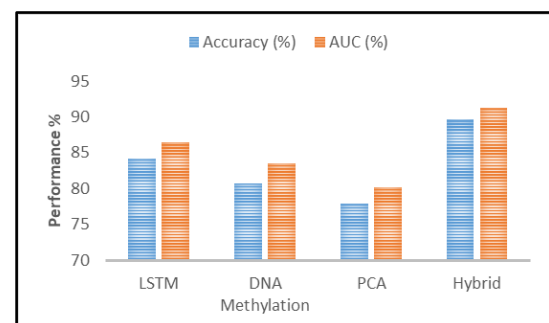


Fig. 3: Performance Comparison of AD Prediction Methods

VII. CONCLUSION&FUTURE WORK

This review has brought out the current scenario in Alzheimer's disease prediction and detection and the challenges ahead. Different techniques that exhibit different investigations are presented in this study. The prediction methods along with their accuracy are studied along with providing insights for detecting the presence of the disease at an early stage. It also gives a brief summary on the works that have been conducted previously by various authors in the area of Machine Learning and Deep Learning. However, the importance of early detection and prediction of the disease needs much more work to be done. Advancement in optimization techniques and integrating them, interpreting the models and improving their accuracy and scarcity of data still remain challenging. The studies and results achieved in the previous works will be motivational for further research in this area.

REFERENCES

- [1]. E. M. Reiman, J. B. Langbaum, and P. N. Tariot, Alzheimer's prevention initiative: A proposal to evaluate presymptomatic treatments as quickly as possible, *Biomarkers Med.*, vol. 4, no. 1, pp. 314, 2010.
- [2]. Jedynak, B. M. et al. A computational neurodegenerative disease progression score: method and results with the Alzheimer's disease neuroimaging initiative cohort. *Neuroimage* 63, 1478–1486 (2012).
- [3]. Young, A. L. et al. A data-driven model of biomarker changes in sporadic Alzheimer's disease. *Brain* 137, 2564–2577 (2014).
- [4]. W. Zaremba, I. Sutskever, and O. Vinyals, Recurrent neural network regularization, 2014, arXiv:1409.2329. [Online]. Available: <https://arxiv.org/abs/1409.2329>.
- [5]. Donohue, M.C.; Sperling, R.A.; Salmon, D.P.; Rentz, D.M.; Raman, R.; Thomas, R.G.; Weiner, M.; Aisen, P.S. The preclinical Alzheimer cognitive composite: Measuring amyloid-related decline. *JAMA Neurol.* 2014, 71, 961–970.
- [6]. S. Rathore, M. Habes, M.A. Iftikhar, A. Shacklett, C. Davatzikos, A review on neuroimaging-based classification studies and associated feature extraction methods for Alzheimer's disease and its prodromal stages, *NeuroImage* 155 (2017) 530–548.
- [7]. R. Katzman, The prevalence and malignancy of Alzheimer disease: A major killer, *Arch. Neurol.*, vol. 33, no. 4, pp. 217218, 1976.
- [8]. World Health Organization, Dementia, 2020. <https://www.who.int/news-room/fact-sheets/detail/dementia>.
- [9]. Olivari, B.S.; Baumgart, M.; Taylor, C.A.; McGuire, L.C. Population measures of subjective cognitive decline: A means of advancing public health policy to address cognitive health. *Alzheimer's Dement. Transl. Res. Clin. Interv.* 2021, 7, e12142.
- [10]. Siddhartha Kumar Arjaria, Abhishek Singh Rathore, Dhananjay Bisen, Sanjib Bhattacharyya, 2022, Performances of Machine Learning Models for Diagnosis of Alzheimer's Disease.
- [11]. Liu, S. et al. Generalizable deep learning model for early Alzheimer's disease detection from structural MRIS. *Sci. Rep.* 12, 17106. <https://doi.org/10.1038/s41598-022-20674-x> (2022).
- [12]. Xin Hong, Rongjie Lin, Chenhui Yang, Nianyin Zeng, Chunting Cai, Jin Gou, And Jane Yang, 2019, Predicting Alzheimer's Disease Using LSTM, pp. 80893-80901.
- [13]. Gerardo Alfonso Perez and Javier Caballero Villarraso, 2021, Alzheimer Identification through DNA Methylation and Artificial Intelligence Techniques.
- [14]. M. Sudharsan, G. Thailambal, 2021, Alzheimer's disease prediction using machine learning techniques and principal component analysis (PCA).
- [15]. Jinhua Sheng, Bocheng Wanga, Qiao Zhang, Rougang Zhou, Luyun Wang, Yu Xin, 2021, Identifying and characterizing different stage toward Alzheimer's disease using ordered core features and machine learning.